ASSIGNMENT: DEPLOYMENT – DOCKER, FASTAPI, INFERENCE

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Aim: To build and deploy a Machine Learning model using FastAPI, containerize it using Docker, and host it publicly for inference.

Tech Stack Used:

Ubuntu (Development Environment)

FastAPI (API creation)

Docker (Containerization)

Docker Hub (Image Registry)

Render (Cloud Hosting)

Postman (API Testing)

Routes:

FastAPI Development

Created endpoints:

/health → to check API status

/predict → to get predictions

Objective

To build, containerize, and deploy a Machine Learning model using FastAPI and Docker, and make it accessible via a public URL.

Learning Outcomes

Learned how to build a REST API using FastAPI for machine learning inference

Understood how to containerize an application using Docker

Gained knowledge of Docker image creation, tagging, and pushing to Docker Hub

Learned how to deploy a containerized application on a cloud platform (Render)

Understood how to test APIs using Postman and Swagger UI

Learned how to debug common errors such as 404 routes and input validation issues

Understood the importance of correct ports and environment configuration in deployment

Gained practical experience in end-to-end ML deployment workflow

Architecture Overview

```
Local Machine (Ubuntu)          Cloud Platform (Docker Hub + Render)
-----------------------------------------------------------------------

train_model.py                  Docker Hub (Container Registry)
     |                          |
     v                          v
app/iris_model.pkl              [iris-api:v1 image]
     |                          |
     v                          v
app/main.py   -- docker build -->    Render Web Service
Dockerfile    -- docker push  -->         |
                                          v
                                  Public HTTPS Endpoint
                                https://iris-api-v1-lxkd.onrender.com/predict
```

Module 0: Prerequisites & Setup

Steps Followed :

Trained Iris classification model using scikit-learn

Saved model as .pkl file

Installing Reason:I used "sudo apt install python3 python3-pip -y" to install Python and pip on Ubuntu.

Python3 is required to run the FastAPI application.

python3-pip is needed to install required libraries like fastapi and uvicorn.

`sudo` allows executing commands with administrator privileges.

`apt` is the package manager used to install software.

The `-y` flag automatically confirms installation without asking for user input.

```
→  Docker_Fast_api
→  Docker_Fast_api sudo apt install python3 python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.10.6-1~22.04.1).
python3 set to manually installed.
python3-pip is already the newest version (22.0.2+dfsg-1ubuntu0.7).
0 upgraded, 0 newly installed, 0 to remove and 52 not upgraded.
→  Docker_Fast_api ps -p 1778
    PID TTY          TIME CMD
```

Module 1: Model Training

```
touch train_model.py
touch requirements.txt
touch Dockerfile
→  iris-deploy echo "fastapi==0.111.0" >> requirements.txt
echo "uvicorn[standard]==0.29.0" >> requirements.txt
echo "scikit-learn==1.4.2" >> requirements.txt
echo "numpy==1.26.4" >> requirements.txt
echo "pydantic==2.7.1" >> requirements.txt
→  iris-deploy pip3 install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Collecting fastapi==0.111.0
  Downloading fastapi-0.111.0-py3-none-any.whl (91 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 92.0/92.0 KB 1.4 MB/s eta 0:00:00
Collecting uvicorn[standard]==0.29.0
  Downloading uvicorn-0.29.0-py3-none-any.whl (60 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 60.8/60.8 KB 6.2 MB/s eta 0:00:00
```

I created the app folder using mkdir app in Ubuntu to store the trained model file.

I trained the machine learning model and saved it as iris_model.pkl using pickle.

The accuracy is printed to verify that the model is working correctly.

Then I created main.py to build the FastAPI application.

I defined input and output classes to handle request and response data.

Finally, I added API endpoints like /health and /predict to check status and make predictions.

```
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /home/yasaswini/.local/lib/python3.1
>=13.7.1->rich-toolkit>=0.14.8->fastapi-cli>=0.0.2->fastapi==0.111.0->-r requirements.txt (line
Requirement already satisfied: mdurl~=0.1 in /home/yasaswini/.local/lib/python3.10/site-package
.2.0->rich>=13.7.1->rich-toolkit>=0.14.8->fastapi-cli>=0.0.2->fastapi==0.111.0->-r requirements
→  iris-deploy python3 train_model.py
→  iris-deploy
→  iris-deploy python3 train_model.py
→  iris-deploy echo "from sklearn.datasets import load_iris" >> train_model.py
echo "from sklearn.ensemble import RandomForestClassifier" >> train_model.py
echo "from sklearn.model_selection import train_test_split" >> train_model.py
echo "import pickle" >> train_model.py
echo "import os" >> train_model.py
echo "" >> train_model.py
echo "iris=load_iris()" >> train_model.py
```

```
Select ../iris-deploy
echo "os.makedirs('app',exist_ok=True)" >> train_model.py
echo "with open('app/iris_model.pkl','wb') as f:" >> train_model.py
echo "    pickle.dump(model,f)" >> train_model.py
echo "" >> train_model.py
echo "print('Model saved to app/iris_model.pkl')" >> train_model.py
→ iris-deploy python3 train_model.py
Model accuracy: 100.00%
Model saved to app/iris_model.pkl
→ iris-deploy >....
echo "CLASS_NAMES=['setosa','versicolor','virginica']" >> app/main.py
echo "" >> app/main.py
echo "class IrisInput(BaseModel):" >> app/main.py
echo "    sepal_length:float" >> app/main.py
echo "    sepal_width:float" >> app/main.py
echo "    petal_length:float" >> app/main.py
echo "    petal_width:float" >> app/main.py
echo "" >> app/main.py
echo "class PredictionOutput(BaseModel):" >> app/main.py
echo "    species:str" >> app/main.py
echo "    species_id:int" >> app/main.py
echo "    confidence:float" >> app/main.py
echo "" >> app/main.py
echo "@app.get('/health')" >> app/main.py
echo "def health():" >> app/main.py
echo "    return {'status':'healthy'}" >> app/main.py
echo "" >> app/main.py
echo "@app.post('/predict',response_model=PredictionOutput)" >> app/main.py
echo "def predict(input:IrisInput):" >> app/main.py
echo "    try:" >> app/main.py
echo "        features=np.array([[input.sepal_length,input.sepal_width,input.petal_length,input.petal_width]])" >> app/m
```

Module 2: FastAPI Application Development

```
→ iris-deploy python3 -m uvicorn app.main:app --reload --port 8000
INFO:     Will watch for changes in these directories: ['/home/yasaswini/calibo/Docker_Fast_api
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [2685] using WatchFiles
INFO:     Started server process [2687]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:42604 - "GET / HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:42604 - "GET / HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:42604 - "GET /demo HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:42604 - "GET / HTTP/1.1" 404 Not Found
INFO:     127.0.0.1:42616 - "GET /health HTTP/1.1" 200 OK
INFO:     127.0.0.1:34406 - "GET /predict HTTP/1.1" 405 Method Not Allowed
INFO:     127.0.0.1:48834 - "GET /predict HTTP/1.1" 405 Method Not Allowed
INFO:     127.0.0.1:58988 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:58988 - "GET /openapi.json HTTP/1.1" 200 OK
```

404 Error

- Cause: Wrong route used
- Fix: Use /health or /docs

```
→ iris-deploy python3 -m uvicorn app.main:app --reload --port 8000
INFO:     Will watch for changes in these directories: ['/home/yasaswini/calibo/Docker_Fast_api/iris-dep
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [2438] using WatchFiles
INFO:     Started server process [2440]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

127.0.0.1:8000

127.0.0.1:8000

Pretty-print

{"detail":"Not Found"}

Module 3: Local Testing

Go to proper valid routes

Use valid routes `/health` or `/docs`; otherwise, a 404 'Path not found' error occurs.

FastAPI - Swagger UI

127.0.0.1:8000/docs

Summarise

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET /health Health

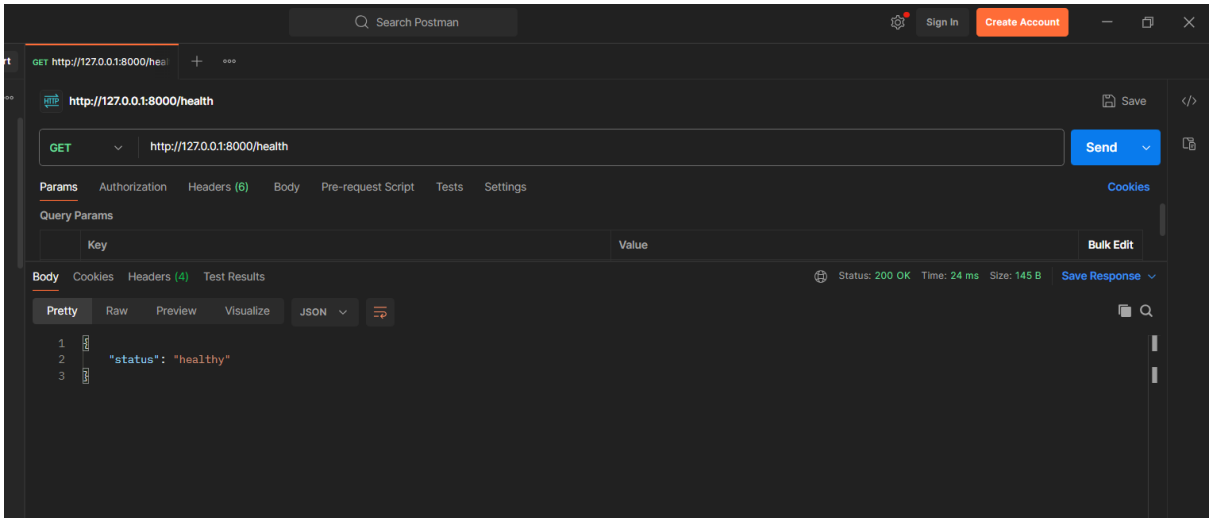
POST /predict Predict

127.0.0.1:8000/health

127.0.0.1:8000/health

Pretty-print

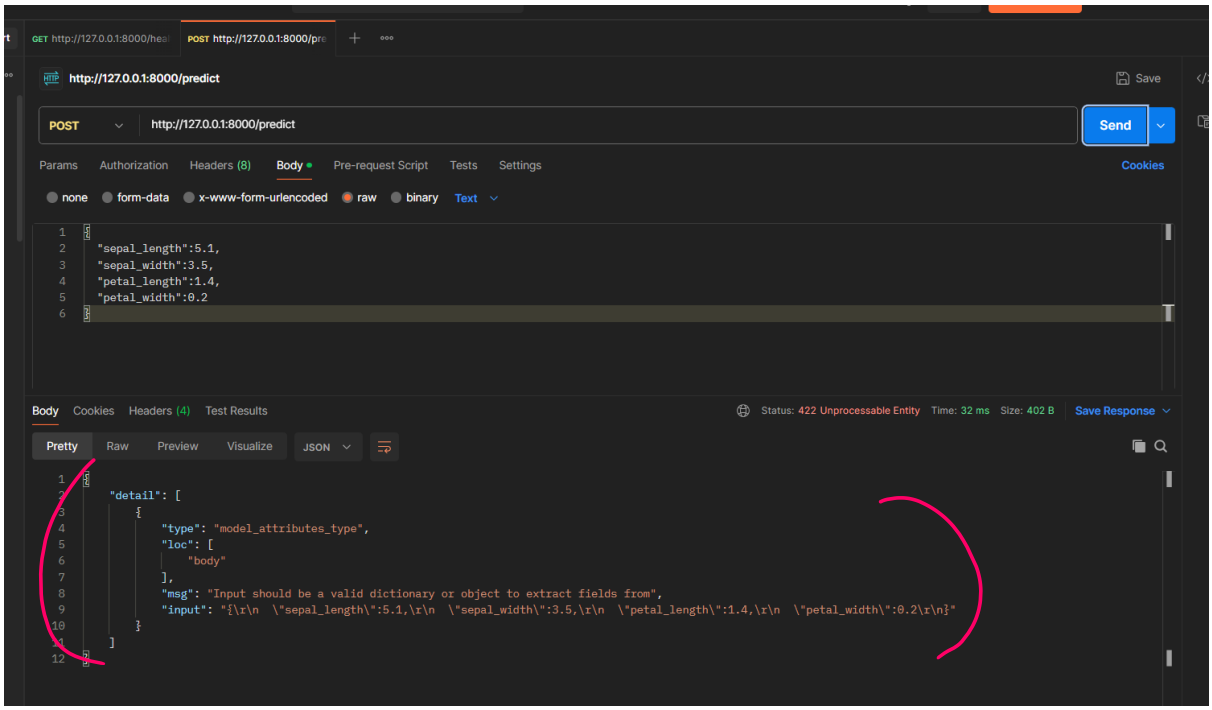
{"status":"healthy"}



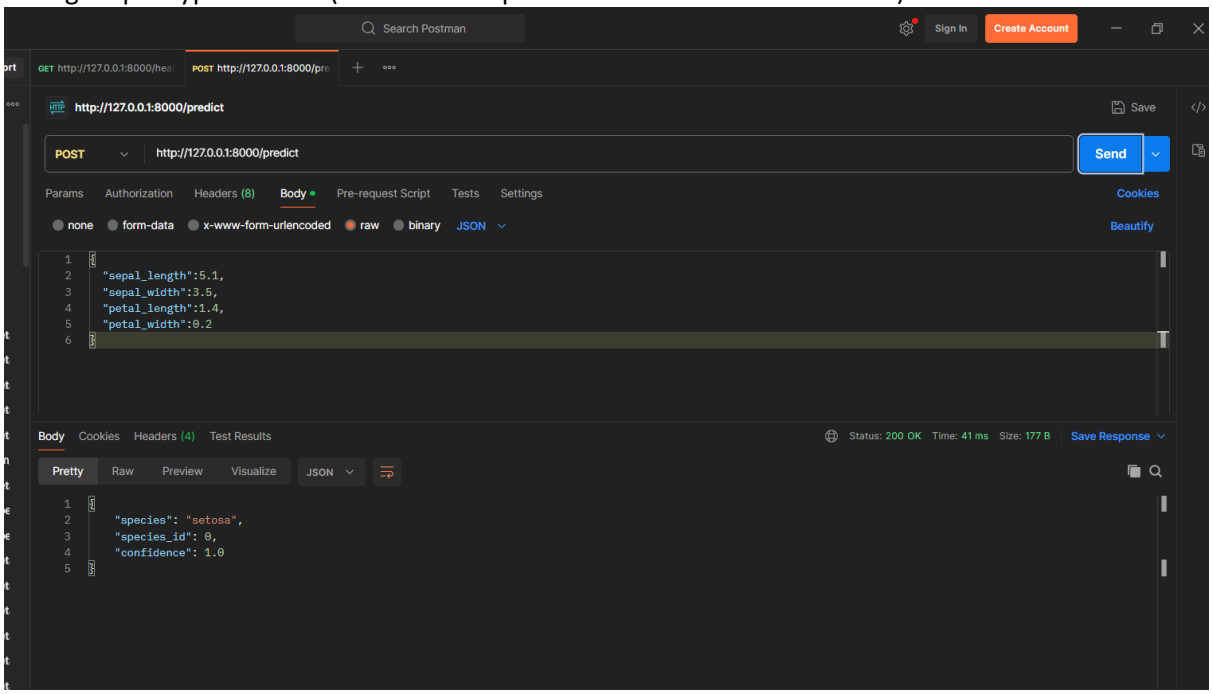
Post input error

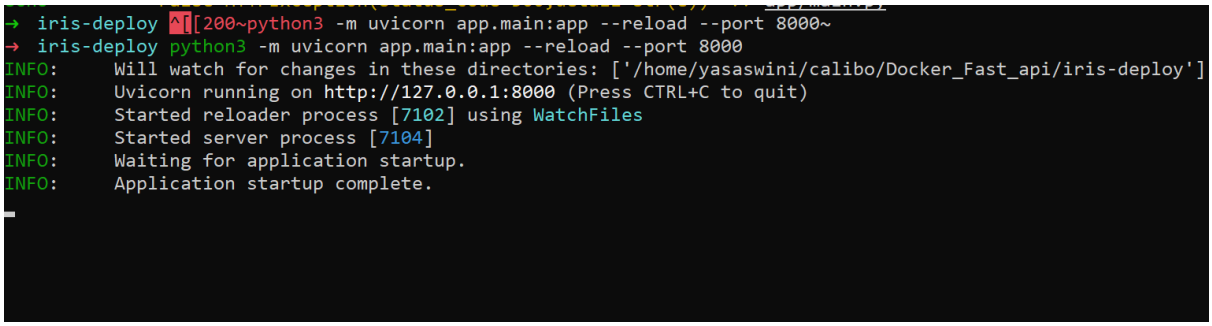
Input Format Error

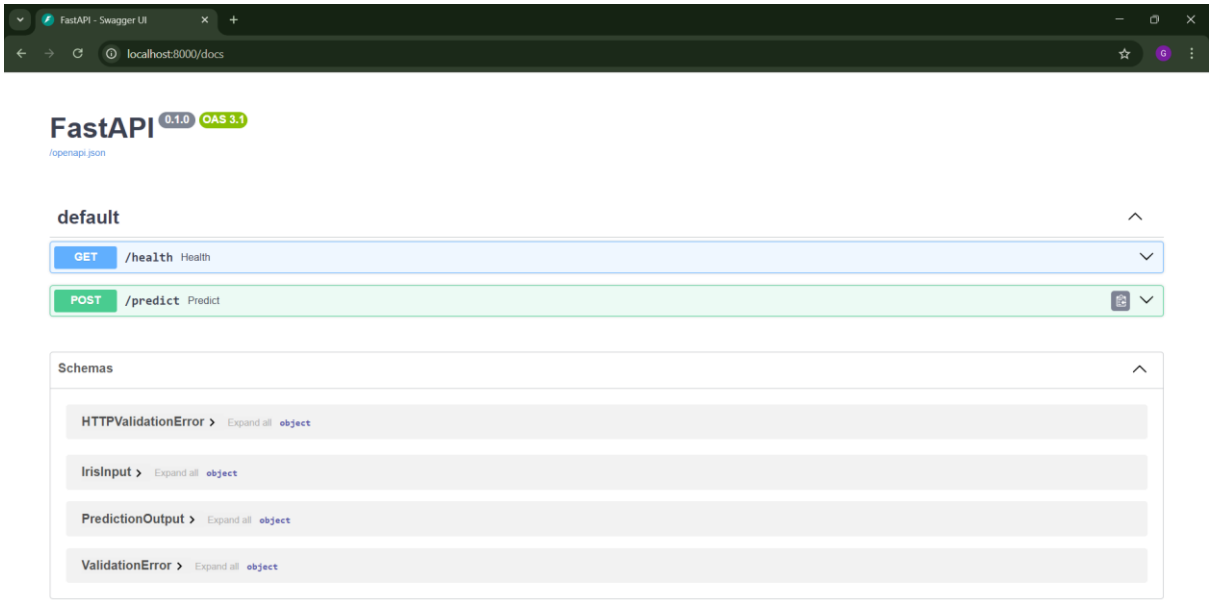
Cause: Incorrect JSON format



Change input type to JSON.(Ensure the input is in the correct JSON format.)







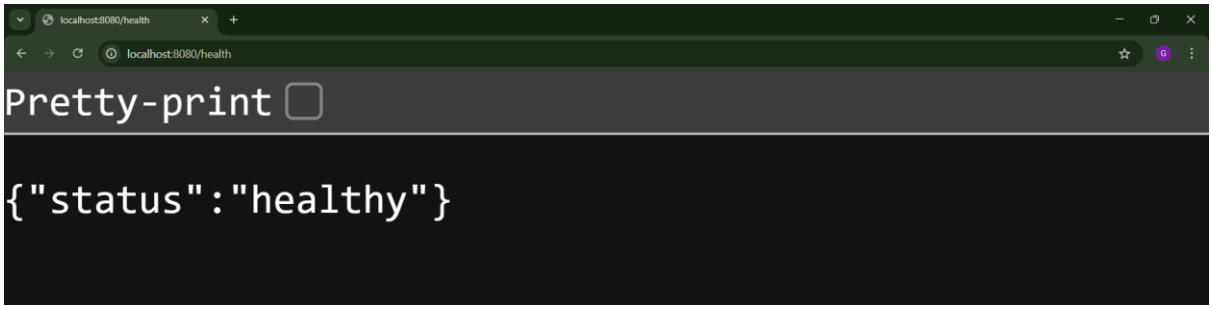
Module 4: Docker Containerization

Next: Docker Steps

```
../iris-deploy
echo "        raise HTTPException(status_code=500,detail=str(e))" >> app/main.py
→ iris-deploy ^[[200~python3 -m uvicorn app.main:app --reload --port 8000~
→ iris-deploy python3 -m uvicorn app.main:app --reload --port 8000
INFO:     Will watch for changes in these directories: ['/home/yasaswini/calibo/Docker_Fast_api/iris-deploy']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [7102] using WatchFiles
INFO:     Started server process [7104]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:54014 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:54014 - "GET /openapi.json HTTP/1.1" 200 OK
^CINFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [7104]
INFO:     Stopping reloader process [7102]
→ iris-deploy
→ iris-deploy echo "FROM python:3.11-slim" >> Dockerfile
echo "" >> Dockerfile
echo "WORKDIR /app" >> Dockerfile
echo "" >> Dockerfile
echo "COPY requirements.txt ." >> Dockerfile
echo "RUN pip install --no-cache-dir -r requirements.txt" >> Dockerfile
echo "" >> Dockerfile
echo "COPY app/ ./app/" >> Dockerfile
echo "" >> Dockerfile
echo "EXPOSE 8080" >> Dockerfile
echo "" >> Dockerfile
echo "CMD [\"uvicorn\",\"app.main:app\",\"--host\",\"0.0.0.0\",\"--port\",\"8080\"]" >> Dockerfile
→ iris-deploy
```

```
→ iris-deploy docker build -t iris-api:v1 .
failed to fetch metadata: fork/exec /usr/local/lib/docker/cli-plugins/docker-buildx: no such file or directory

DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon    190kB
Step 1/7 : FROM python:3.11-slim
3.11-slim: Pulling from library/python
206356c42440: Pulling fs layer
206709444425: Pulling fs layer
bbf418262cbe: Pulling fs layer
43272e9c9d4b: Pulling fs layer
43272e9c9d4b: Waiting
206709444425: Verifying Checksum
206709444425: Download complete
43272e9c9d4b: Verifying Checksum
43272e9c9d4b: Download complete
```

```
 ---> Removed intermediate container 9fbac16c0489
 ---> 5a840def6532
Step 5/7 : COPY app/ ./app/
 ---> 5dd2518c6af5
Step 6/7 : EXPOSE 8080
 ---> Running in 330478ecc3e0
 ---> Removed intermediate container 330478ecc3e0
 ---> a4834e38bd16
Step 7/7 : CMD ["uvicorn","app.main:app","--host","0.0.0.0","--port","8080"]
 ---> Running in 7ad4a27d3423
 ---> Removed intermediate container 7ad4a27d3423
 ---> aa7328baa48a
Successfully built aa7328baa48a
Successfully tagged iris-api:v1
→ iris-deploy docker run -p 8080:8080 iris-api:v1
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```



Error and Debugging

```
Aborted!
→ iris-deploy curl -X POST http://localhost:8080/predict \
-H "Content-Type: application/json" \
-d '{"sepal_length":6.3,"sepal_width":3.3,"petal_length":6.0,"petal_width":2.5}'
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Connection refused
→ iris-deploy curl http://localhost:8080/health
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Connection refused
→ iris-deploy
```

```
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Connection refused
→ iris-deploy curl http://localhost:8080/health
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Connection refused
→ iris-deploy docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS          PORTS
                NAMES
a867f3ad7dc3   mongo:latest   "docker-entrypoint.s…"   6 months ago   Up 15 minutes   0.0.0.0:27017->27017/tcp, [::]:270
17->27017/tcp   mongd_db
→ iris-deploy
```

Testing While Container Running

- Cause: Terminal blocked
- Fix: Open another terminal

```
                    mystifying_dubinsky
→ iris-deploy docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
     PORTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                    NAMES
3bef393feb5b   iris-api:v1                          "uvicorn app.main:ap…"   2 minutes ago   Exited (1) About a minute
ago
                    frosty_lehmann
d7d3342542e4   gcr.io/k8s-minikube/kicbase:v0.0.48   "/usr/local/bin/entr…"   3 months ago    Exited (255) 3 months ago
     127.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.
1:32772->32443/tcp   minikube
7c57847f9435   nginx:latest                          "/docker-entrypoint.…"   6 months ago    Exited (255) 5 months ago
     0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
                    my_nginx
a867f3ad7dc3   mongo:latest                          "docker-entrypoint.s…"   6 months ago    Up 15 minutes
     0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
                    mongd_db
a59f75c505c3   my-apache                             "apache2ctl -D FOREG…"   6 months ago    Created

                    apache-container
c326072fb0d0   ubuntu                                "bash -c 'apt-get up…"   6 months ago    Exited (255) 6 months ago
     0.0.0.0:80->80/tcp, [::]:80->80/tcp
                    my-apache
e93f5935cbcd   hello-world                           "/hello"                 7 months ago    Exited (0) 7 months ago

                    mystifying_dubinsky
→ iris-deploy
```

```
                    mystifying_dubinsky

→ iris-deploy docker logs 3bef393feb5b
NFO:     Started server process [1]
NFO:     Waiting for application startup.
NFO:     Application startup complete.
NFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
NFO:     172.17.0.1:44222 - "GET /docs HTTP/1.1" 200 OK
NFO:     172.17.0.1:44222 - "GET /openapi.json HTTP/1.1" 200 OK
NFO:     172.17.0.1:44218 - "GET /health HTTP/1.1" 200 OK
NFO:     172.17.0.1:44218 - "GET /favicon.ico HTTP/1.1" 404 Not Found
NFO:     Shutting down
NFO:     Waiting for application shutdown.
NFO:     Application shutdown complete.
NFO:     Finished server process [1]

borted!
→ iris-deploy
```

Solution: **Open another terminal in parallel without stopping the running container.**

Now test the API.

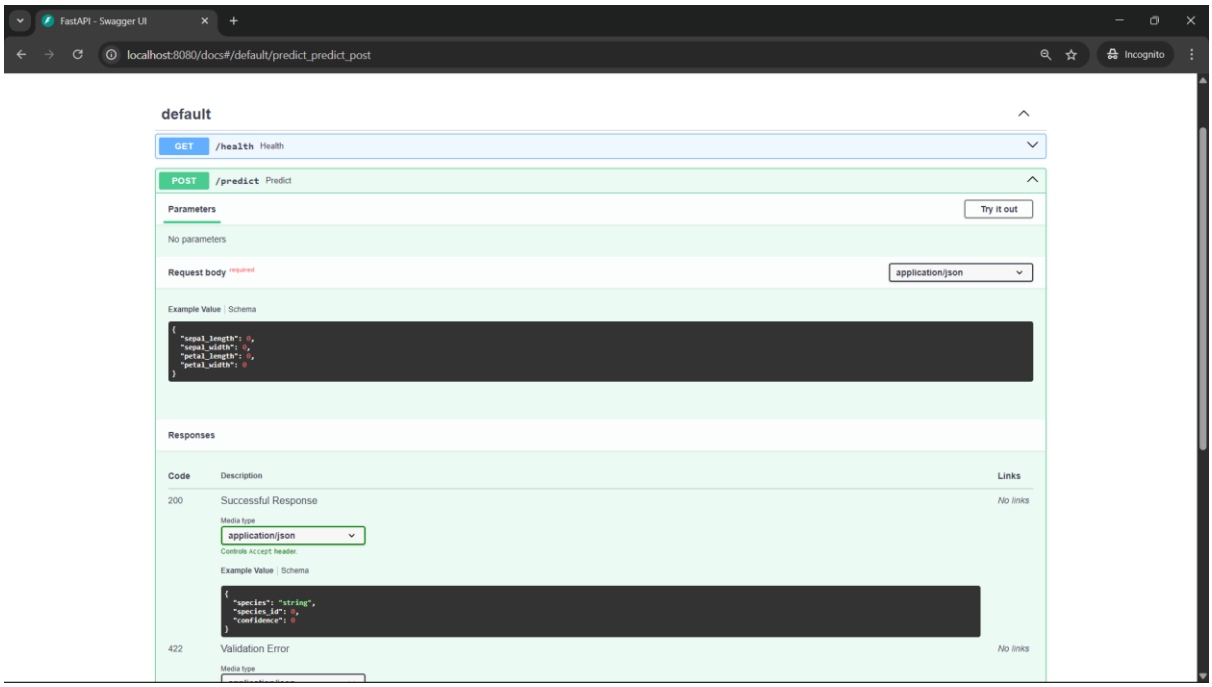
Health:

```
                    mystifying_dubinsky
→ iris-deploy docker logs 3bef393feb5b
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:44222 - "GET /docs HTTP/1.1" 200 OK
INFO:     172.17.0.1:44222 - "GET /openapi.json HTTP/1.1" 200 OK
INFO:     172.17.0.1:44218 - "GET /health HTTP/1.1" 200 OK
INFO:     172.17.0.1:44218 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [1]

Aborted!
→ iris-deploy docker run -p 8080:8080 iris-api:v1
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:55494 - "GET /health HTTP/1.1" 200 OK
```

```
~
→ ~ curl http://localhost:8080/health
{"status":"healthy"}%
→ ~
```

```
                    my-apache
e93f5935cbcd   hello-world                    "/hello"                 7 months ago    Exited (0) 7 months ago

                    mystifying_dubinsky
→ iris-deploy docker logs 3bef393feb5b
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:44222 - "GET /docs HTTP/1.1" 200 OK
INFO:     172.17.0.1:44222 - "GET /openapi.json HTTP/1.1" 200 OK
INFO:     172.17.0.1:44218 - "GET /health HTTP/1.1" 200 OK
INFO:     172.17.0.1:44218 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [1]

Aborted!
→ iris-deploy docker run -p 8080:8080 iris-api:v1
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:55494 - "GET /health HTTP/1.1" 200 OK
INFO:     172.17.0.1:50654 - "POST /predict HTTP/1.1" 200 OK
INFO:     172.17.0.1:34256 - "POST /predict HTTP/1.1" 200 OK
```

```
~
→ ~ curl http://localhost:8080/health
{"status":"healthy"}%
→ ~ curl -X POST http://localhost:8080/predict \
-H "Content-Type: application/json" \
-d '{"sepal_length":6.3,"sepal_width":3.3,"petal_length":6.0,"petal_width":2.5}'
{"species":"virginica","species_id":2,"confidence":1.0}%
→ ~ curl -X POST http://localhost:8080/predict \
-H "Content-Type: application/json" \
-d '{"sepal_length":6.3,"sepal_width":3.3,"petal_length":6.0,"petal_width":2.5}'
{"species":"virginica","species_id":2,"confidence":1.0}%
→ ~
```

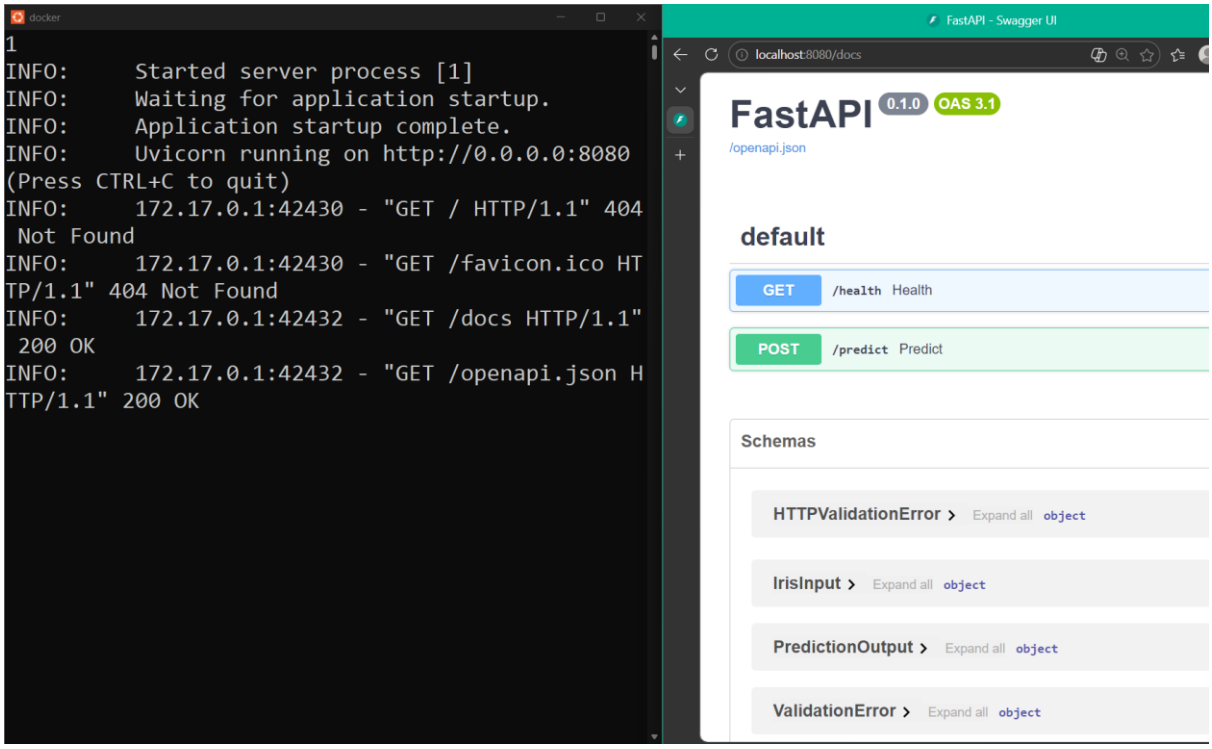


```
→ ~ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED         STATUS          PORTS
                 NAMES
3a719ff40cf1   iris-api:v1    "uvicorn app.main:ap…"   2 minutes ago   Up 2 minutes    0.0.0.0:8080->8080/tcp, [::]:8080
->8080/tcp       relaxed_kepler
a867f3ad7dc3   mongo:latest   "docker-entrypoint.s…"   6 months ago    Up 19 minutes   0.0.0.0:27017->27017/tcp, [::]:27
017->27017/tcp   mongd_db
→ ~
```

Using docker desktop and hub

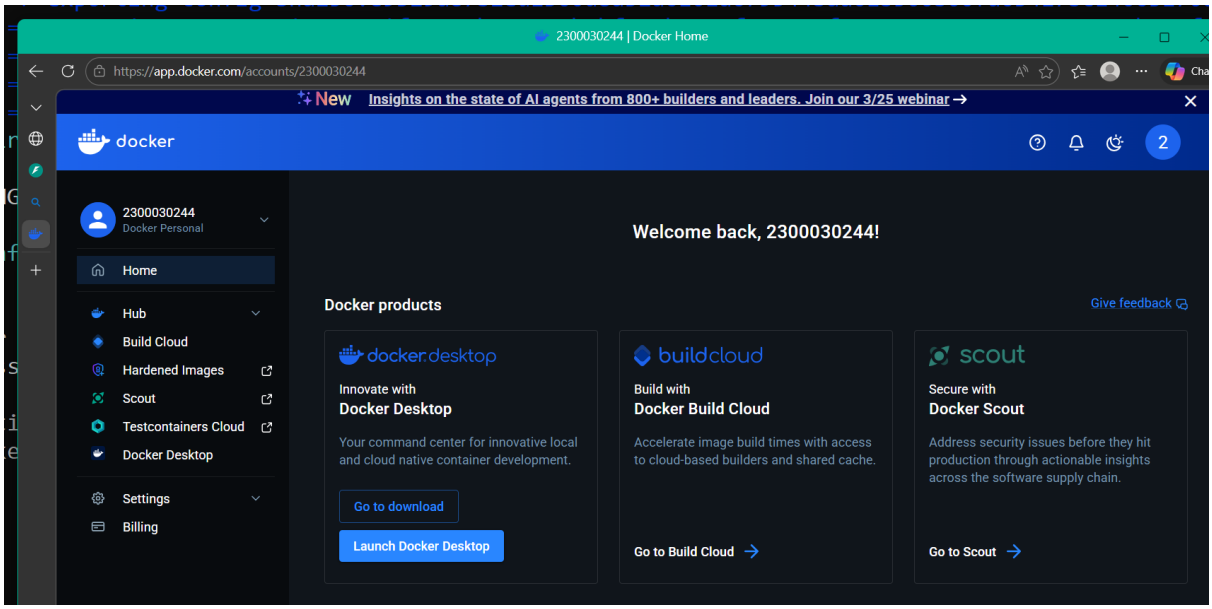
```
Dockerfile  app  requirements.txt  train_model.py
→ iris-deploy docker build -t iris-api:v1 .
[+] Building 59.7s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 249B
 => [internal] load metadata for docker.io/library/python:3.11-slim
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:124d73aee6fbab06658af5fe54d5
 => => resolve docker.io/library/python:3.11-slim@sha256:124d73aee6fbab06658af5fe54d5
 => => sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d1f8f 29.78M
 => => sha256:6944007a5b45fe41656cd72b1691b1419bf9d39641e7afbc937f3a3abd07e575 1.29MB
 => => sha256:7524aba74b8cc45ff3cf2f089fd9a638dfd58b001763194d0819c8d7c75d598f 250B /
 => => sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e226a88 14.37M
 => => extracting sha256:ec781dee3f4719c2ca0dd9e73cb1d4ed834ed1a406495eb6e44b6dfaad5d
 => => extracting sha256:6944007a5b45fe41656cd72b1691b1419bf9d39641e7afbc937f3a3abd07
 => => extracting sha256:e35b92e2619d5152fbb45e6ccb02dbb1007c5784ad19a5e235bf05ca6e22
 => => extracting sha256:7524aba74b8cc45ff3cf2f089fd9a638dfd58b001763194d0819c8d7c75d
 => [internal] load build context
 => => transferring context: 180.77kB
 => [2/5] WORKDIR /app
 => [3/5] COPY requirements.txt .
```

```
Login Succeeded
→ iris-deploy docker run -p 8080:8080 iris-api:v1
INFO:     Started server process [1]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
INFO:     172.17.0.1:42430 - "GET / HTTP/1.1" 404 Not Found
INFO:     172.17.0.1:42430 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO:     172.17.0.1:42432 - "GET /docs HTTP/1.1" 200 OK
INFO:     172.17.0.1:42432 - "GET /openapi.json HTTP/1.1" 200 OK
```

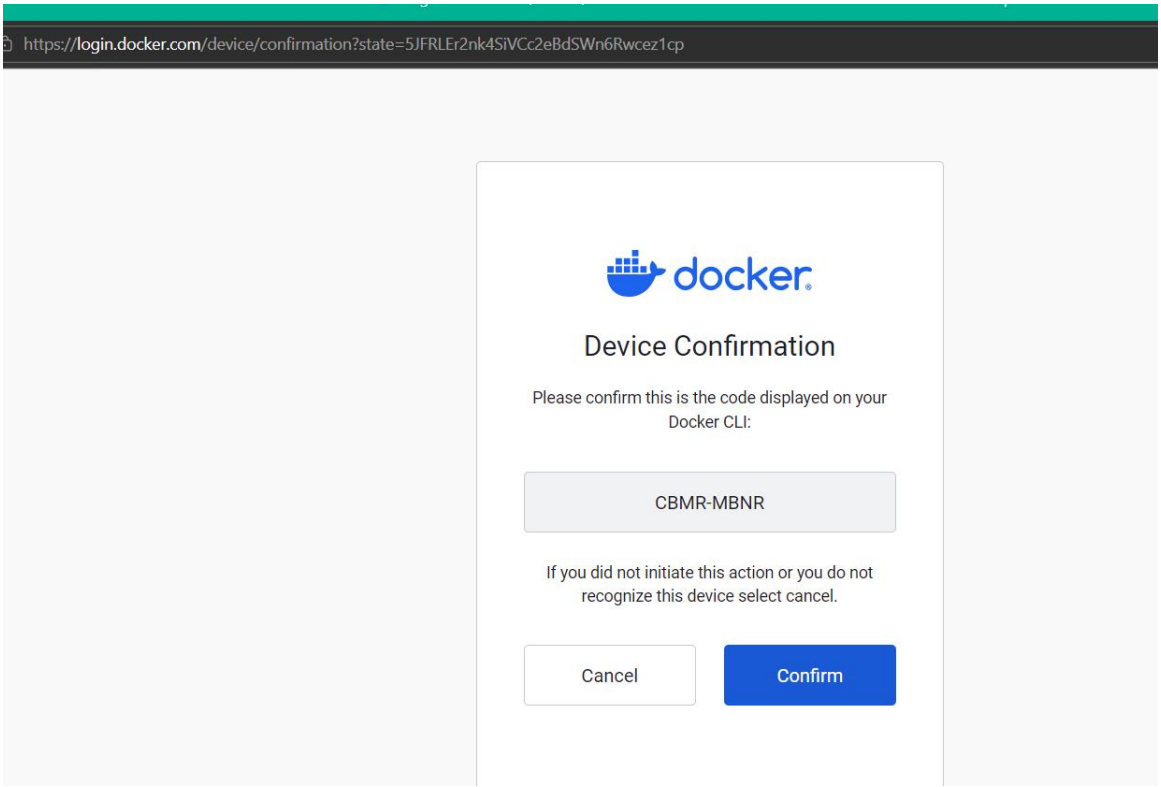


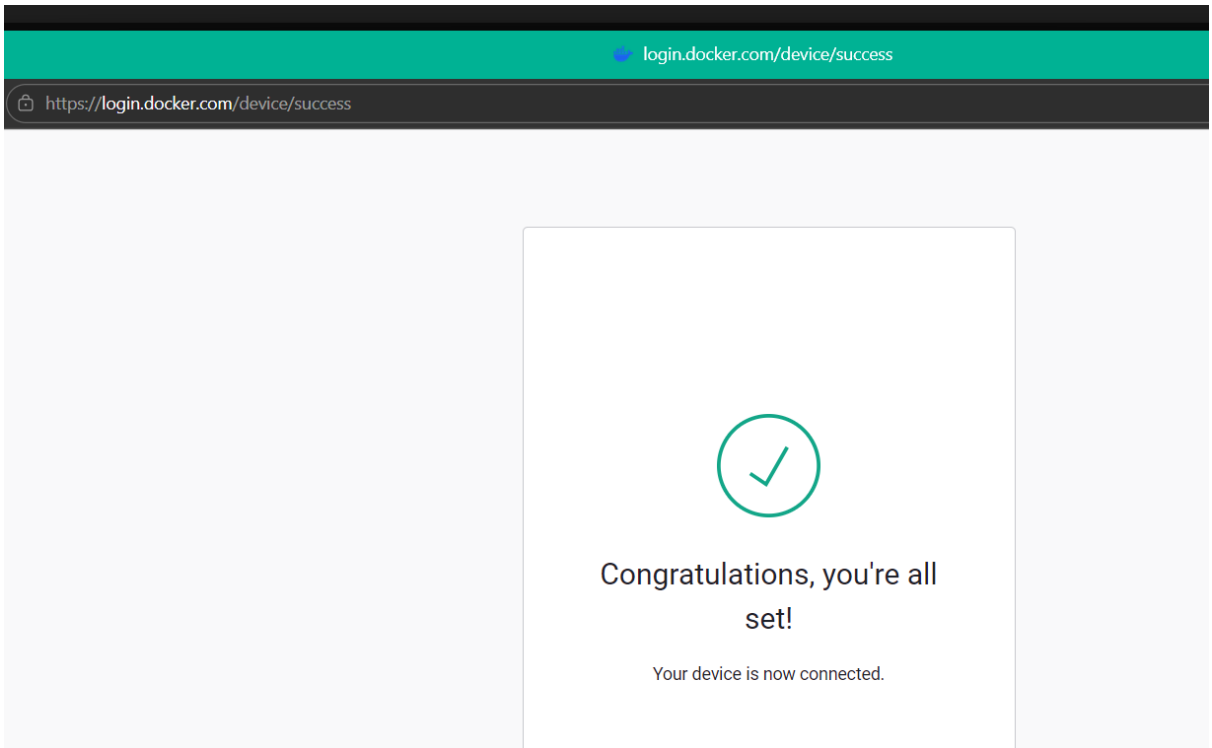
Module 5: Docker Hub Deployment

Login to docker hub in browser and authenticate



```
→ iris-deploy docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: CBMR-MBNR
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…
```





```
→ iris-deploy docker login
Authenticating with existing credentials... [Username: 2300030244]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
→ iris-deploy
```

Instead of Google artifact I am using docker login (Instead of creating a GCP Artifact Registry repository, I am using Docker Hub. I log in using docker login and push my Docker image there)

Login:

docker login

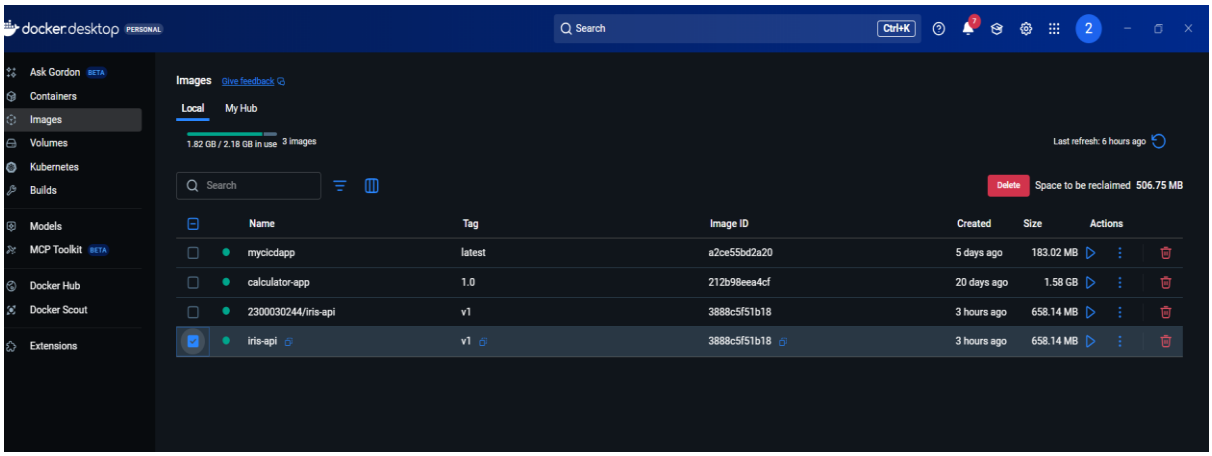
Tag:

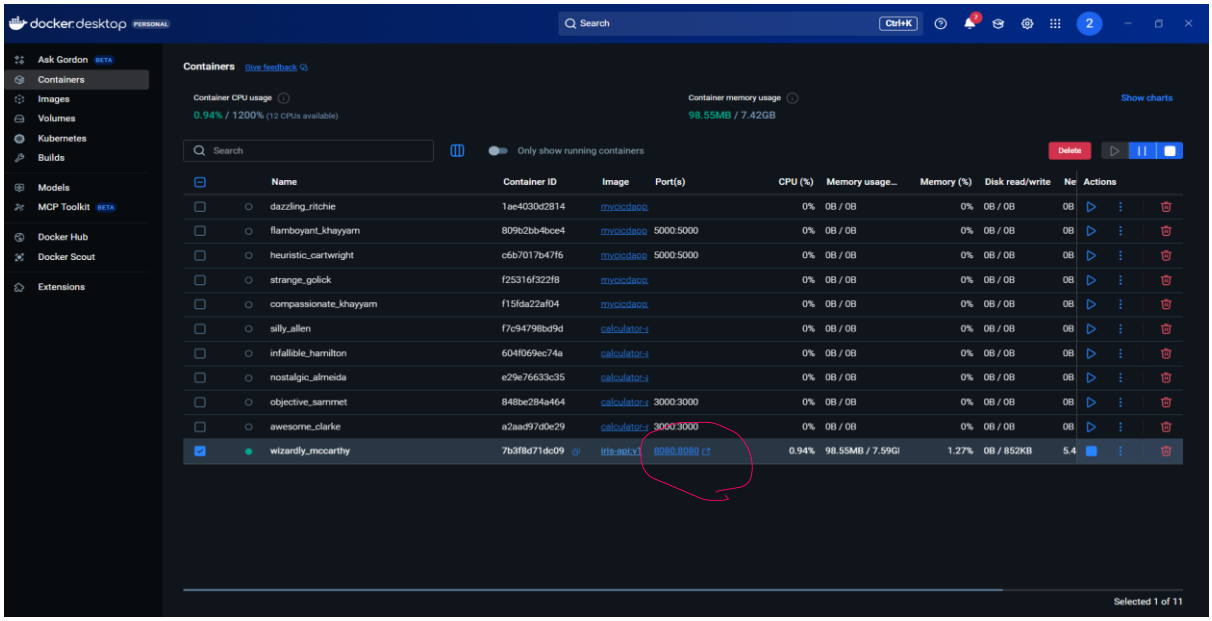
docker tag iris-api:v1 2300030244/iris-api:v1

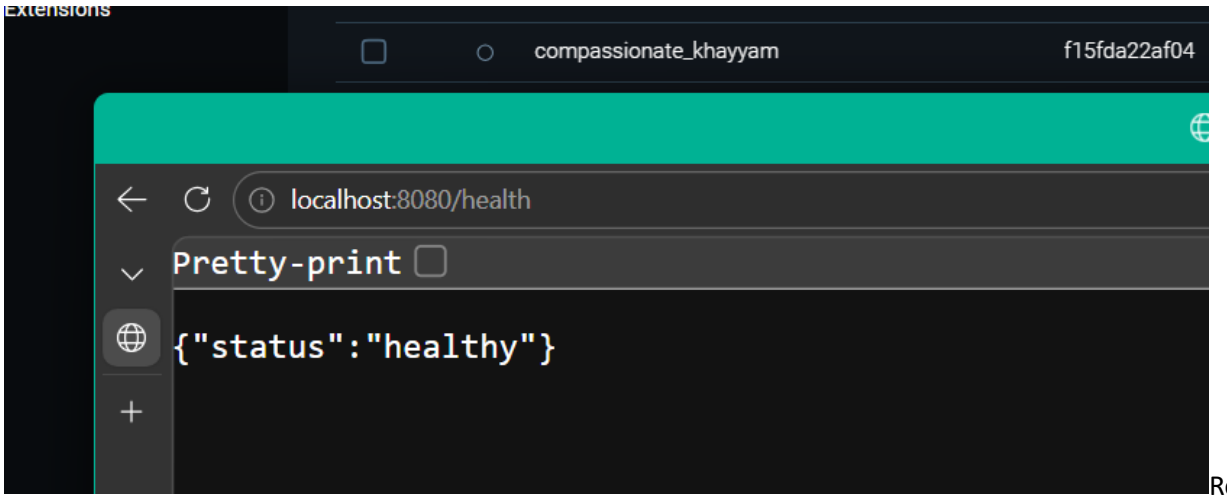
Push:

docker push 2300030244/iris-api:v1

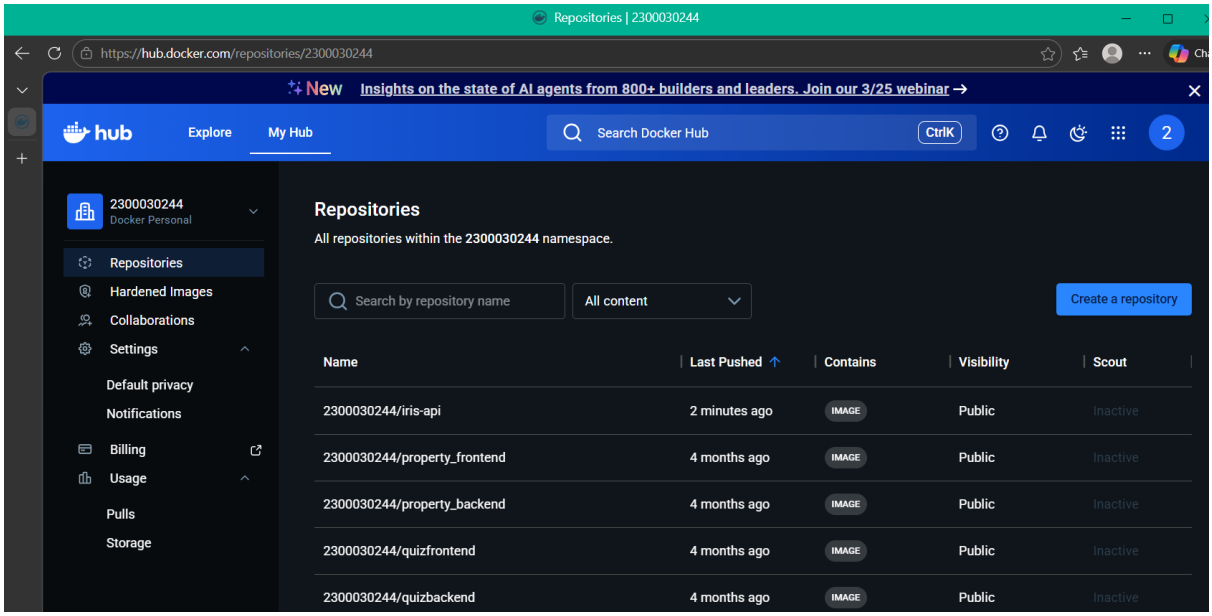
And Render For Any Where access

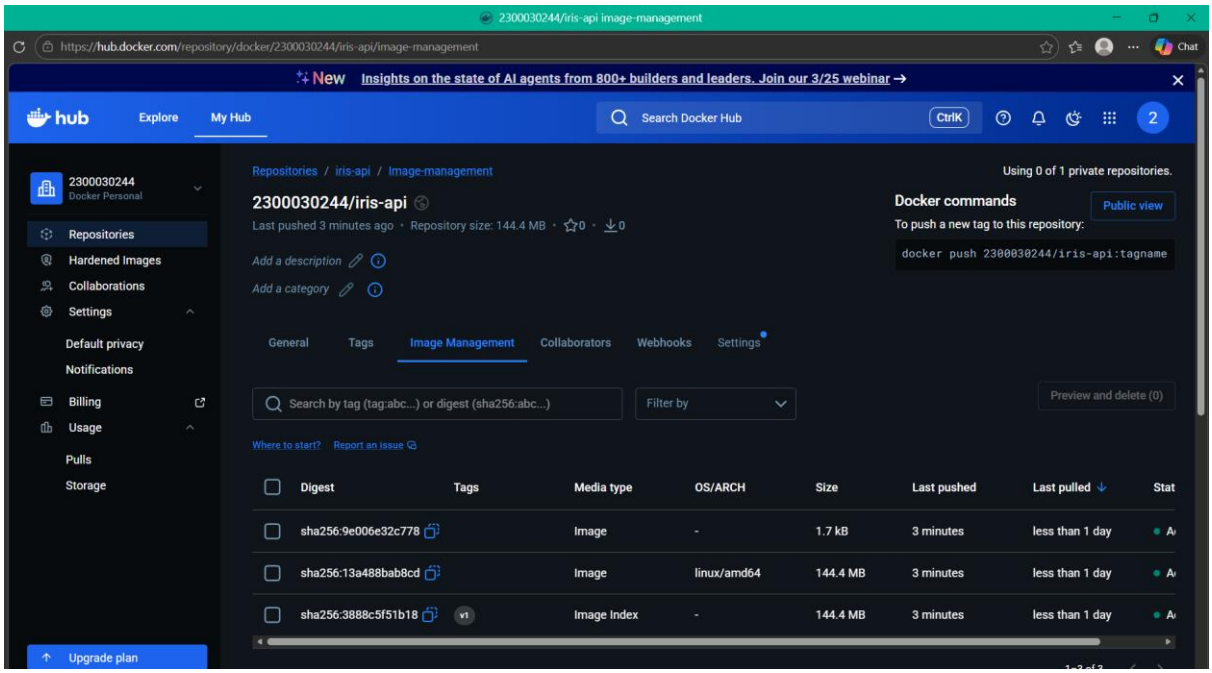


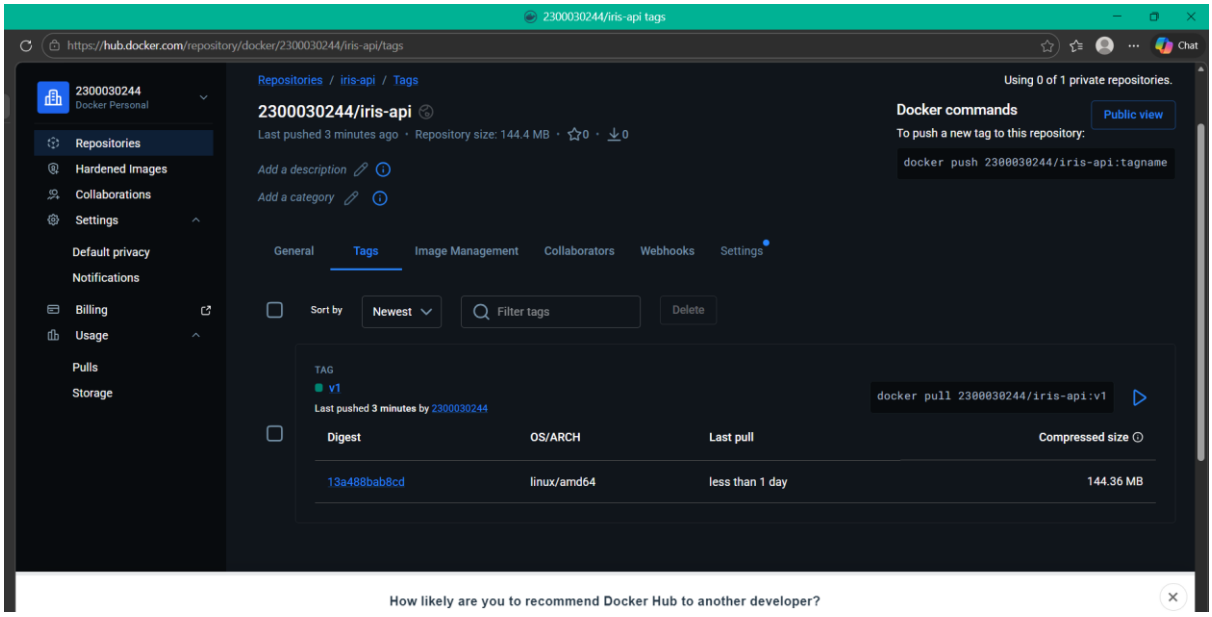


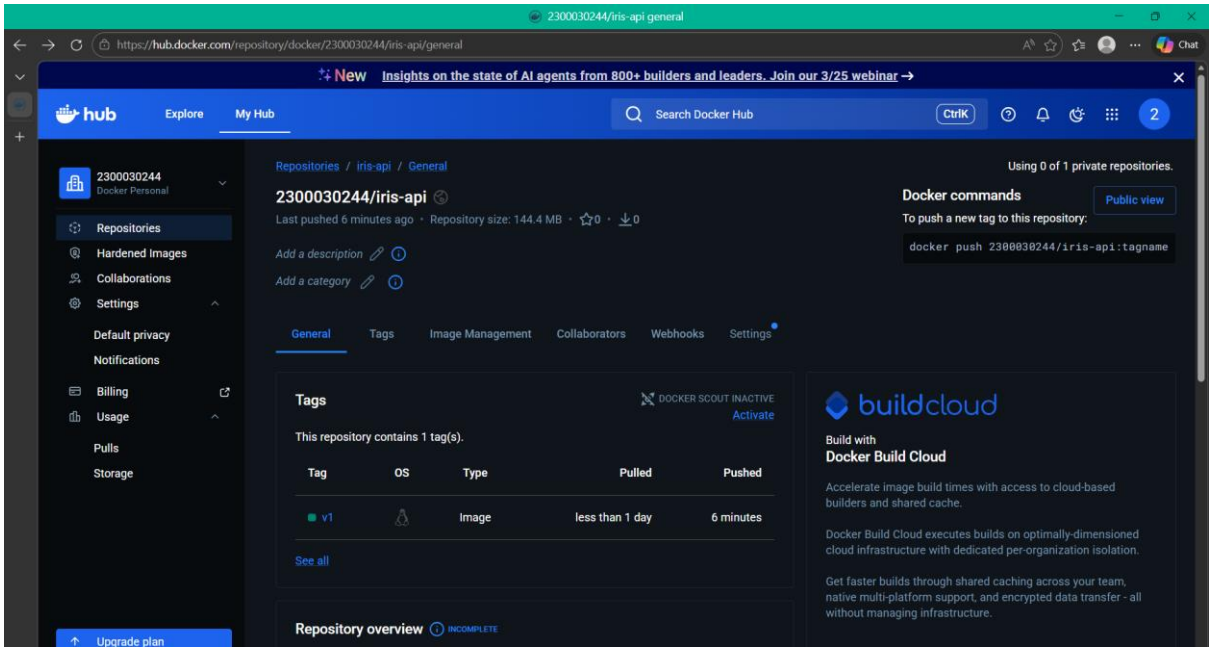


Remote Access of image by docker hub browser login









Module 6: Cloud Deployment (Render)

Now Render for any where Access(Deploying on Render for public access.)

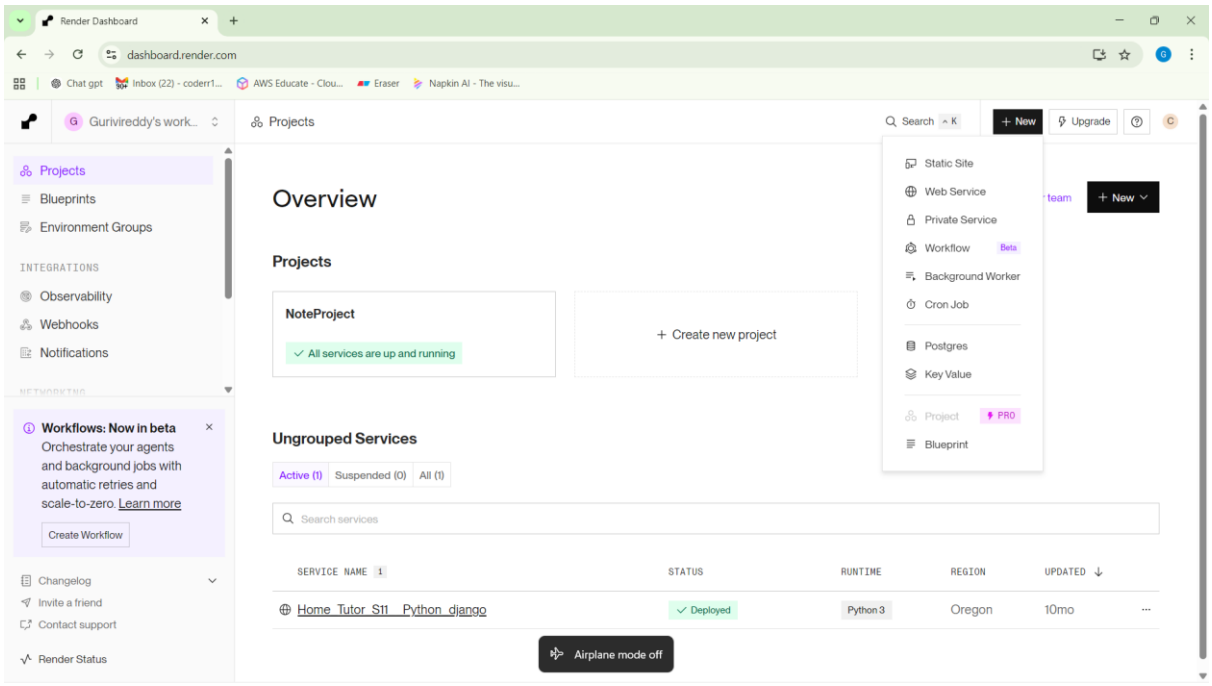
Cloud platforms like Render or GCP Cloud Run are used to host APIs for global access.

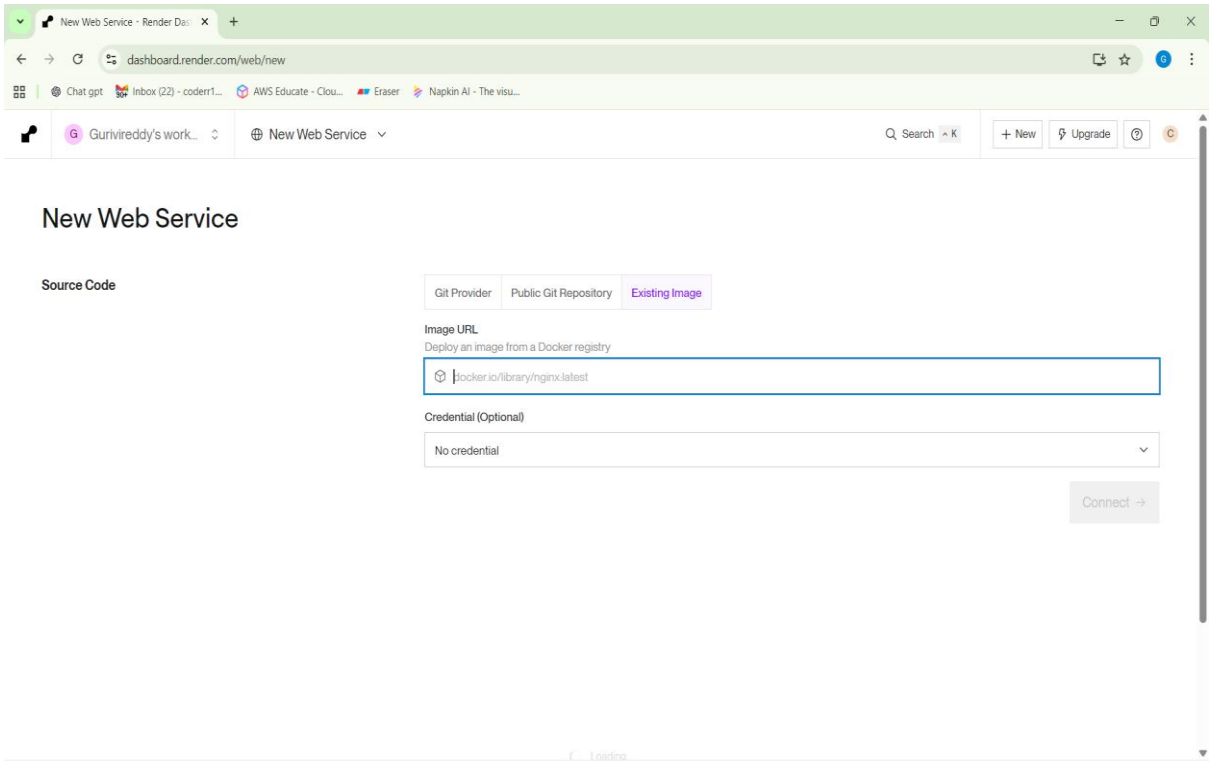
Render Deployment

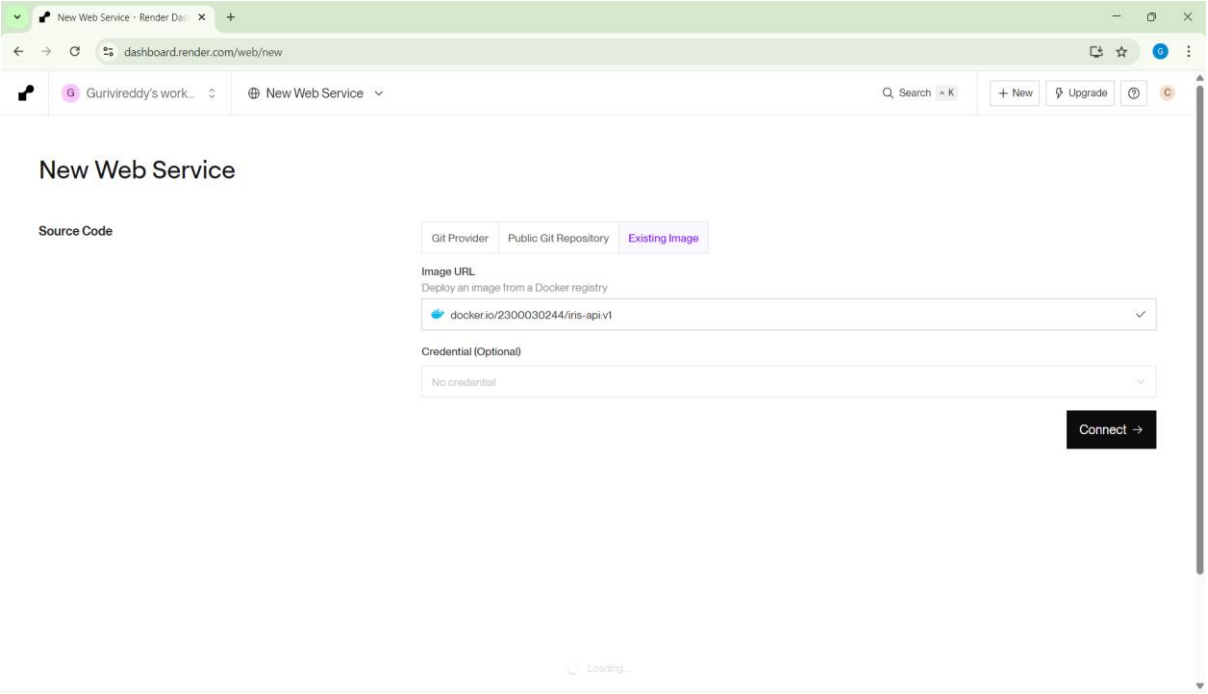
Steps:

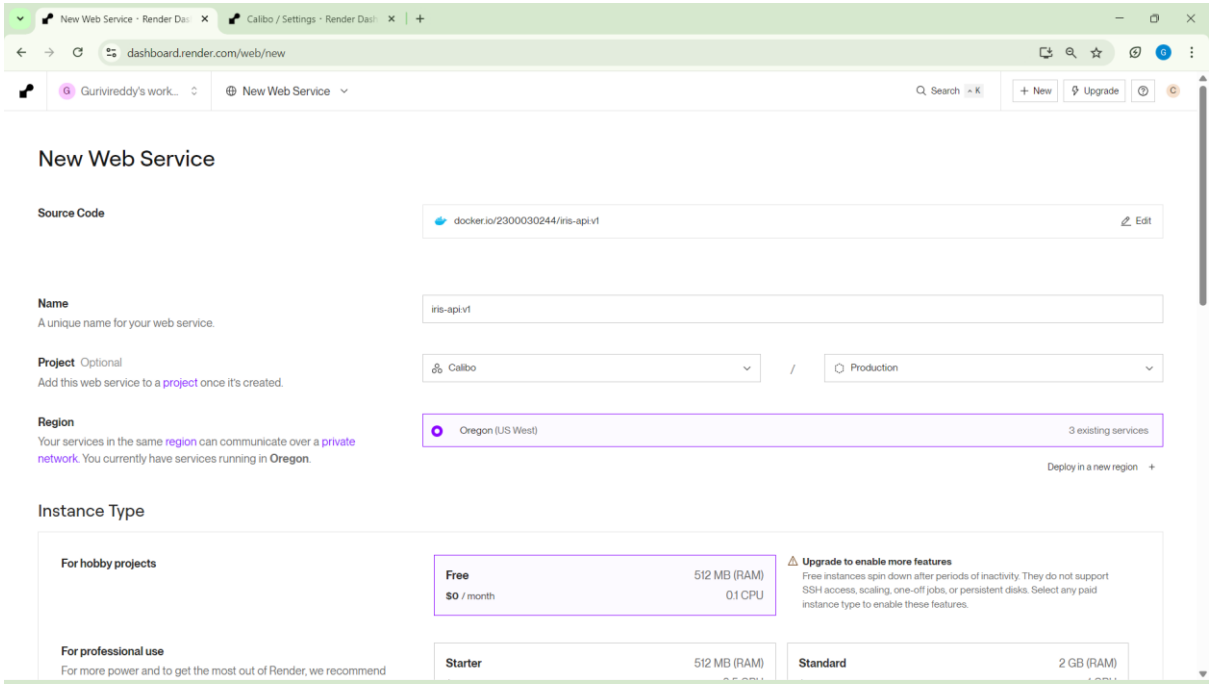
1. Go to Render
2. Click **New +**
3. Select **Web Service**

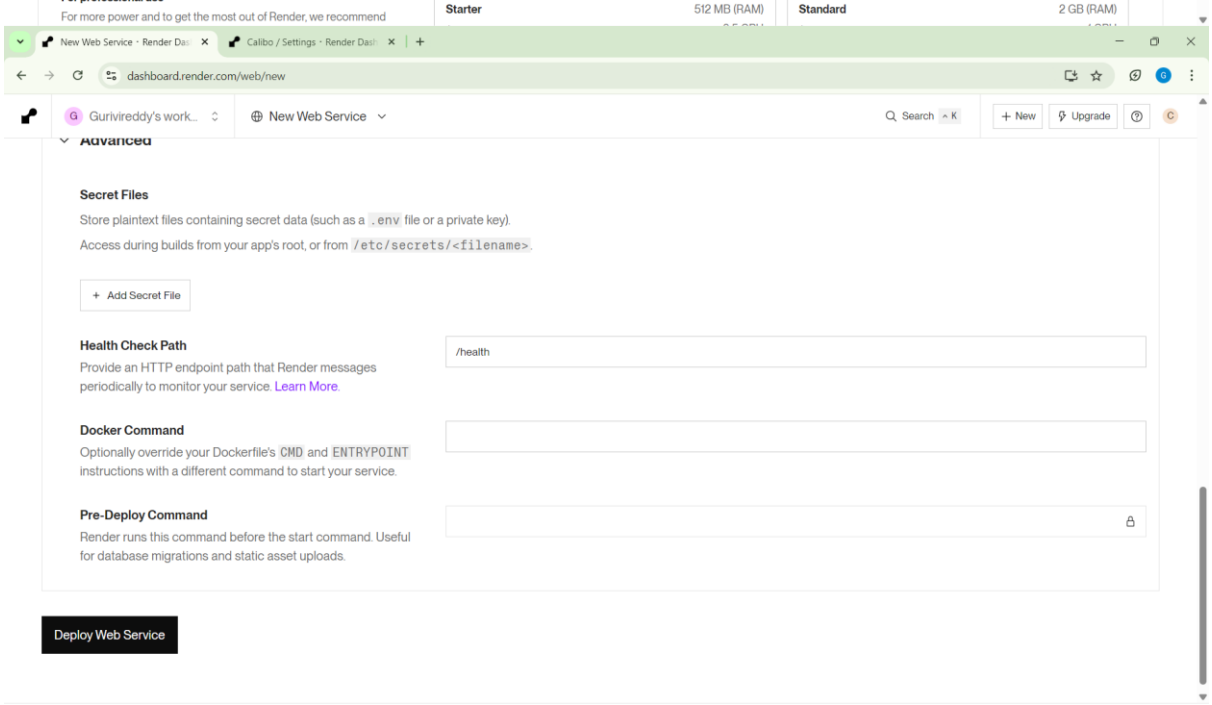
4. Choose **Deploy from existing image**
5. Connect Docker Hub





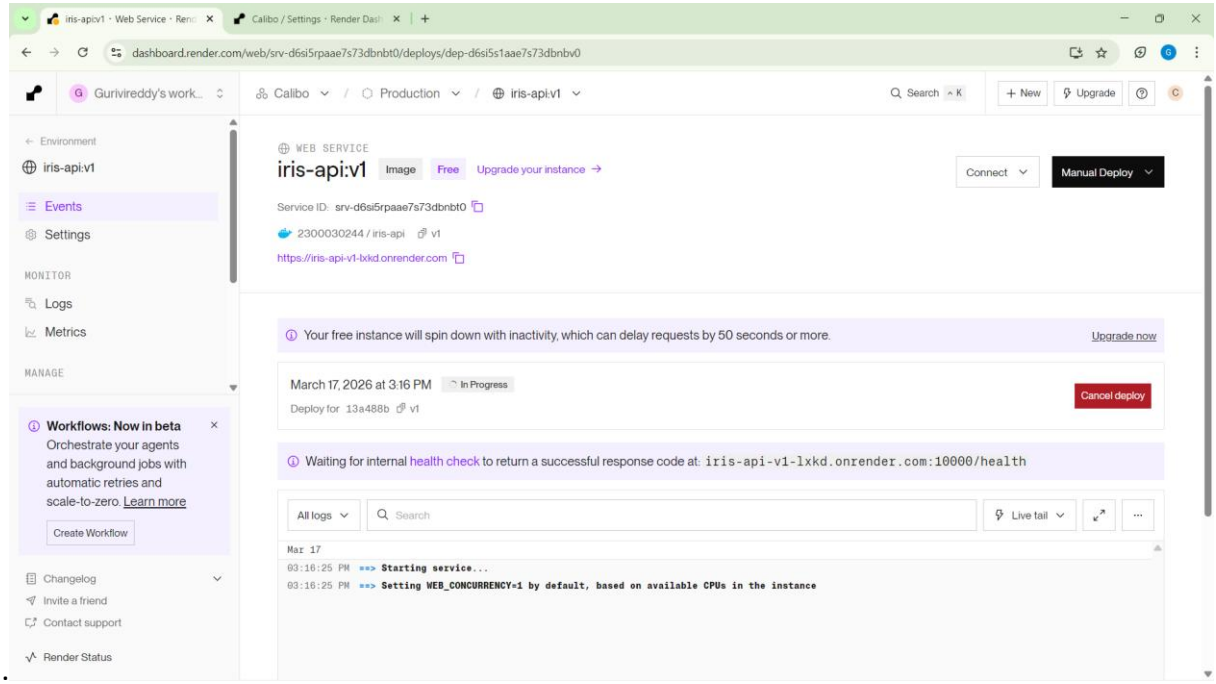

New Web Service · Render Das
dashboard.render.com/web/new
Guriviredddy's work...
New Web Service
Search K
+ New
Upgrade
New Web Service
Source Code
Git Provider
Public Git Repository
Existing Image
Image URL
Deploy an image from a Docker registry
docker.io/2300030244/iris-api:v1
Credential (Optional)
No credential
Connect →
Loading...

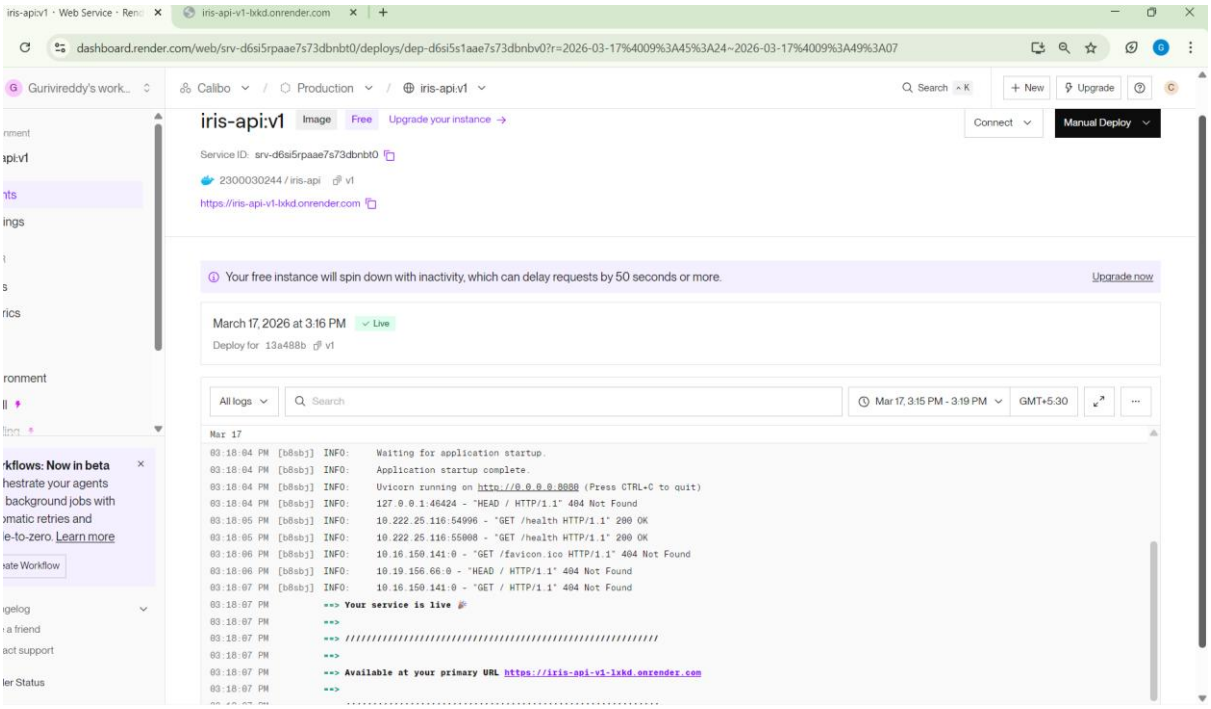


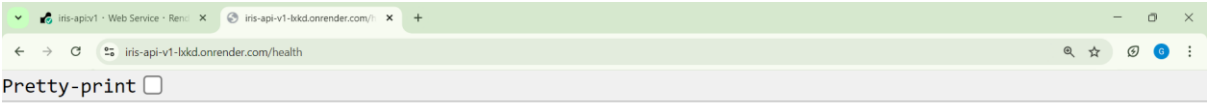


The default Docker CMD instruction is used to start the application.

No need to specify







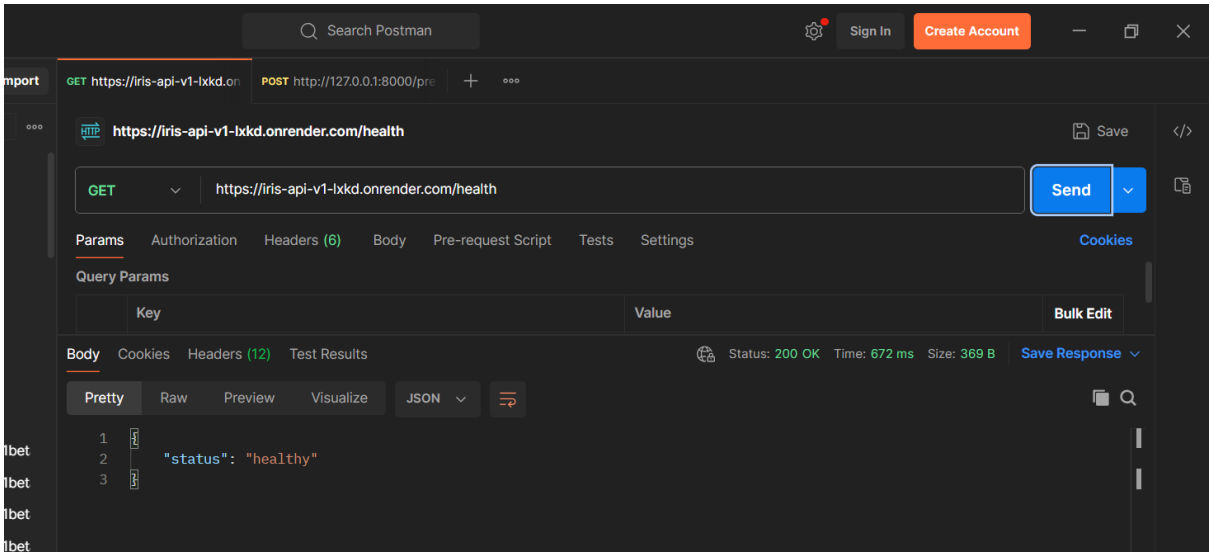
{"status":"healthy"}

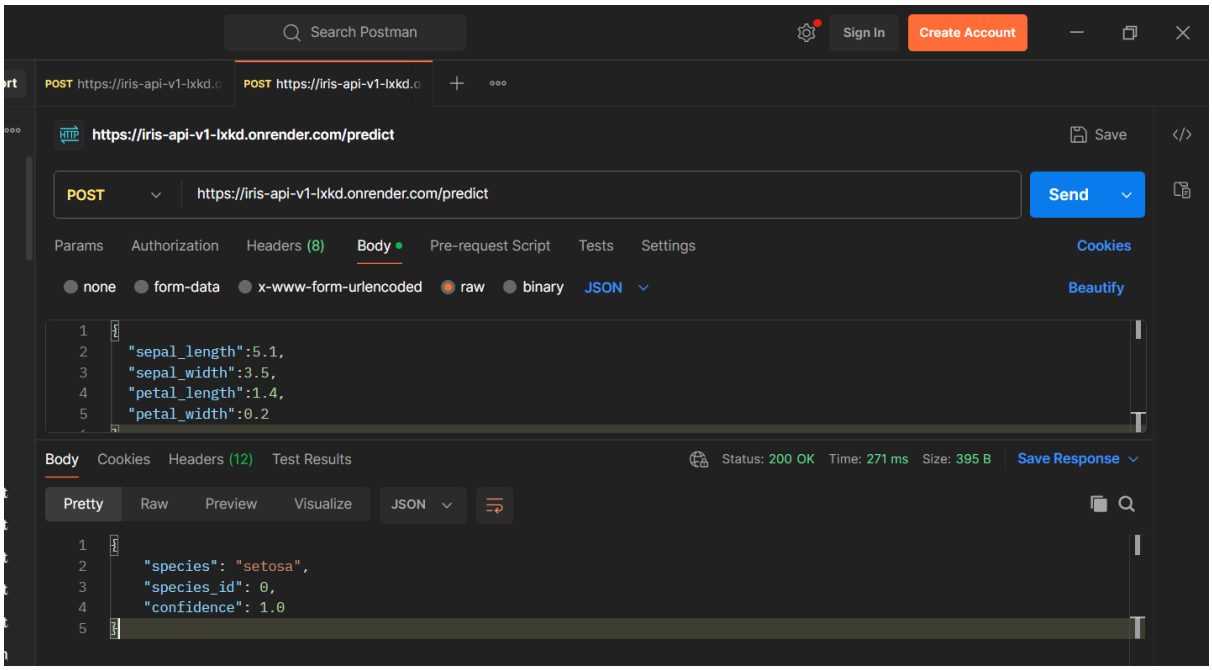
Module 7: API Testing

Testing

Tested using Postman

Verified API responses





Module 9: Deployment Output

LINK : https://iris-api-v1-lxkd.onrender.com

Issues Faced & Debugging: I resolved issues which I have faces in middle of the modules like post data types ,parallel terminal execution..etc

Declaration

I used ChatGPT for guidance and understanding, and implemented the project independently.

Conclusion:

The assignment was successfully implemented using Ubuntu, Docker, Docker Hub, and Render. Even though GCP/Azure was suggested, the deployment objective was achieved using alternative tools..